

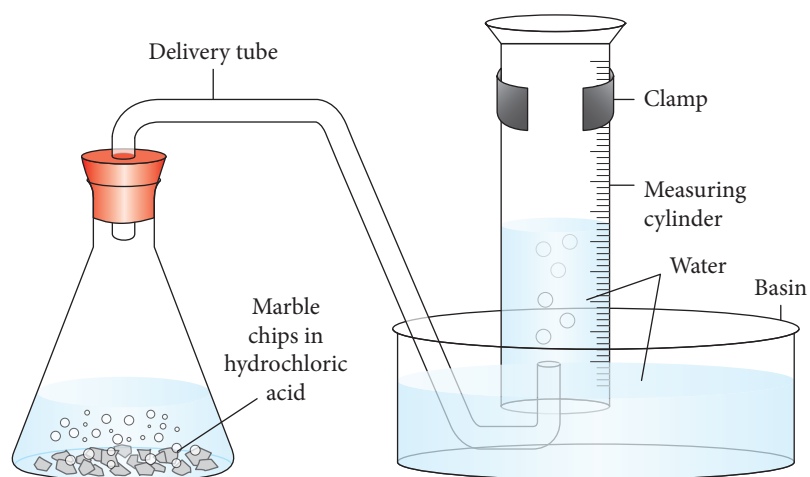
Name: _____

Class: _____

ACTIVITY SHEET

9.5 Graphing and interpreting results

In an experiment, two grams of marble chips (calcium carbonate) were placed in excess dilute hydrochloric acid, and the gas produced in the reaction was measured using the apparatus shown.



1 Explain why the measuring cylinder was inverted and initially full of water.

2 Describe how you would confirm that the gas produced is carbon dioxide.

The apparatus was used to measure the gas produced each minute over a 10-minute period, then repeated using 2 g of powdered CaCO_3 and fresh dilute excess HCl . The results are shown in the table.

Time (min)	Volume gas (mL)	
	Using chips	Using powder
0	0	0
1	20	70
2	60	130
3	120	200
4	190	280
5	250	370
6	310	430
7	370	460
8	420	470
9	430	470
10	430	470

- 3 State the formulas of the reactants and products.

- 4 Write a word equation and a balanced formula equation for the reaction.

- 5 Plot the reaction progress for the chip and powder on the same graph, using a key.

- 6 Describe the rate of each reaction.
- 7 Identify the reaction that went to completion faster.
- 8 Explain why that reaction went faster than the other one.
- 9 Account for why the faster reaction levelled off earlier.
- 10 Suggest a reason why the amounts of CO_2 produced differed.